EC571: Digital VLSI Circuit Design

Homework 3 – Fall 2026

Total: 100 points
Due: October 11, 2026 at 11:59 pm

General Instructions

Use this worksheet to complete the homework. Please print this worksheet, write your solutions on the worksheet, take a picture of the worksheet, and submit it through Gradescope. If you prefer, you can download the Homework 3 worksheet on your tablet, complete the worksheet on your tablet, and submit the completed worksheet through Gradescope.

Show all reasoning. A final numerical answer without intermediate steps will receive limited credit. For power questions, clearly distinguish **energy per transition**, **average dynamic power**, **short-circuit power**, and **leakage/static power**. For logic-style questions, clearly state which devices conduct for the relevant input cases.

You may use calculators, lecture notes, and lecture slides. You may use AI tools only for concept review. You may not use AI tools to produce final derivations, final numerical answers, final circuit interpretations, or final written explanations. At the end of your submission, include one sentence disclosing whether and how you used AI tools.

Question 1: CMOS Power Accounting in an Activity-Dependent Block [20 points]

A digital block contains three classes of nodes. The supply voltage is $0.9V$, and the clock frequency is $1.2GHz$.

Node class	Number of nodes	Capacitance per node	Activity factor → α
local logic outputs	20,000	3 fF	0.12
medium wires	4,000	18 fF	0.08
long/global wires	600	95 fF	0.03

Assume each listed capacitance is the total capacitance switched by that node. In addition, the block has short-circuit power equal to 8% of the total dynamic switching power under nominal input slopes.

The leakage current per minimum-equivalent transistor is $2.5nA$. The block contains 1.8×10^6 minimum-equivalent transistors.

(a) **Dynamic power by node class.** Compute the dynamic power contributed by each node class. Which class dominates? Explain whether the answer is obvious from capacitance alone. **[6 points]**

$V_{DD} = 0.9V$; $f = 1.2GHz$. $P_{dynamic} = \alpha C V_{DD}^2 f$

1) local logic outputs: $C_{total} = 20000\,(3fF) = 60000fF = 60pF$

$\therefore P_{dynamic\ local} = 0.12(60 \times 10^{-12})(0.9)^2(1.2 \times 10^9)$

$\approx 6.998\ mW$

2) Med. wires: $C_{total} = 4000\,(18fF) = 72pF$

$P_{dynamic\ med.} = 0.08(72 \times 10^{-12})(0.9)^2(1.2 \times 10^9)$

$\approx 5.60\ mW$

3) long / global wires: $C_{total} = 600\,(95fF) = 57pF$

$P_{dynamic\ long} = (0.03)(57 \times 10^{-12})(0.9)^2(1.2 \times 10^9)$

$= 1.662\ mW$

$\therefore$ Local logic outputs dominates. Not obvious from cap. alone bc $C_{med} > C_{logic}$. but activity factor gives more hint bc $0.12 > 0.08 > 0.03$

(b) **Total dynamic and short-circuit power.** Compute total dynamic switching power. Then compute short-circuit power. **[4 points]**

$P_{\text{total dyn.}} = P_{\text{local}} + P_{\text{med}} + P_{\text{long}}$

$= 6.998 + 5.60 + 1.662 \approx \boxed{14.26\ \text{mW}}$

Short-circuit is 8% of total dynamic switching power

$\therefore P_{sc} = 14.26 \times 8\% \approx \boxed{1.14\ \text{mW}}$

(c) **Leakage power.** Compute total leakage power. Compare leakage power against dynamic switching power. **[4 points]**

$I_{\text{total leak}} = I_{\text{leak 1 trans.}} (N_{\text{trans.}})$

$= 2.5\text{nA}\,(1.8 \times 10^6) = \boxed{4.5\ \text{mA}}$

$P_{\text{total leak}} = V_{DD} I_{\text{total leak}}$

$= (0.9\text{V})(4.5\text{mA}) = \boxed{4.05\ \text{mW}}$

Against dynamic switching power:

$\frac{4.05}{14.26} \times 100 \approx \boxed{28.4\%\ \text{dynamic switching power}}$

(d) **Design interpretation.** Suppose a designer reduces the activity of the long/global wires by 50% using clock gating or operand isolation, but this adds 5,000 local logic nodes with activity factor 0.05 and capacitance 3 fF. Does this change reduce total dynamic power? Show your reasoning. **[3 points]**

0.G P_{long} = 1.662 mW → $P_{long\ New}$ = 50%·(1.662) ≈ 0.831 mW

But new overheads from 5k local logic nodes × $\alpha = 0.05$, $C = 3fF$

$P_{overheads} = 5000(0.05)(3\times10^{-5})(0.9)^2(1.2\times10^9)$

$= 0.729$ mW

Net power gained/loss: $\Delta P = 0.729 - 0.831 = -0.102$ mW

Therefore, total dynamic energy decreases

(e) **Reasoning check.** A student says: "The local logic has the most nodes, so it must dominate power." Explain why this statement may be wrong using your numerical results. **[3 points]**

Not always true because $P \propto N\alpha C$ which relies on more than just number of nodes (N).

From 1 a), local logic has 20k nodes with the dominating power at ≈ 7.0 mW. However, medium wires only has 4k nodes but having the power consumption at 5.6 mW (0.8 node ratio while having 5× less amount of nodes)

Question 2: Short-Circuit Current, Input Slope, and Load Dependence [20 points]

Short-circuit current occurs when both pull-up and pull-down networks conduct during an input transition. The short-circuit current waveform depends on the input slope and the load capacitance.

For an inverter, assume the short-circuit current pulse can be approximated as triangular. The supply voltage is $1.0V$.

A simulation produces the following measurements:

Case	Input transition time	Load capacitance	Peak short-circuit current	Short-circuit pulse width
A	20 ps	20 fF	90 μA	18 ps
B	80 ps	20 fF	210 μA	70 ps
C	80 ps	200 fF	95 μA	55 ps

The node switches with activity factor 0.15 at 800 MHz.

(a) **Short-circuit energy.** Compute the short-circuit energy per transition for cases A, B, and C. **[5 points]**

$V_{DD} = 1.0V$. Using triangular approx. for current pulse:

$$Q = \frac{1}{2} I_{peak} t_{pulse} \quad \times E = V_{DD} Q$$

$$\therefore E = \frac{1}{2} V_{DD} I_{peak} t_{pulse}$$

- Case A: $E_A = \frac{1}{2}(1)(90 \times 10^{-6})(18 \times 10^{-12}) = \boxed{0.810\,fJ}$
- Case B: $E_B = \frac{1}{2}(1)(210\,\mu A)(70\,ps) = \boxed{7.35\,fJ}$
- Case C: $E_C = \frac{1}{2}(1)(95\,\mu A)(55\,ps) \approx \boxed{2.61\,fJ}$

(b) **Short-circuit power.** Compute average short-circuit power for cases A, B, and C. **[4 points]**

$P_{sc} = \alpha f E_{sc}$; $\alpha = 0.15$, $f = 800\,MHz$

$\therefore P_{sc} = 0.15(800\,MHz) = 120 \times 10^6$

- Case A : $P_{sc\,A} = 120(10^6)(0.810 \times 10^{-15})$

$= 97.2\,nW$

- Case B : $P_{sc\,B} = 120(10^6)(7.35 \times 10^{-15})$

$= 882\,nW$

- Case C : $P_{sc\,C} = 120(10^6)(2.6125 \times 10^{-15})$

$\approx 313.5\,nW$

(c) **Load dependence.** Compare cases B and C. They have the same input transition time but different load capacitance. Explain why the larger load can reduce the short-circuit current pulse even though it increases capacitive switching energy. **[4 points]**

Case C has larger load capacitance (200fF > 20fF). This means that its voltage output changes more slowly compared to B. During input transition, the voltage across the conducting transistors is reduced, thus lowering the direct VDD → GND short-circuit current pulse.

(d) **Dynamic energy comparison.** For cases A and C, compute the ordinary capacitive switching energy for a rising output transition. Compare the short-circuit energy with the capacitive switching energy in each case. **[4 points]**

$E_{cap} = C_L V_{DD}^2$ (from 0→1 transition)

- Case A : $E_{cap\,A} = 20fF(1)^2 = \boxed{20fJ}$

  vs. switching : $0.81/20 \times 100 = \boxed{4.05\%}$

- Case C : $E_{cap\,C} = 200fF(1)^2 = \boxed{200fJ}$

  vs. switching : $\frac{2.6125}{200} \times 100 = \boxed{1.31\%}$

(e) **Design action.** Give two circuit-level ways to reduce short-circuit power. For each method, state one possible penalty. **[3 points]**

1) Reduce input transition time

↳ Idea: Use a stronger driver so that the input transition faster. This minimizes the time when both pMOS(s) & nMOS(s) are ON.

Penalty: Higher input capacitance.

2) Reduce transistor current

↳ Idea: Use devices with higher $V_{threshold}$ to reduce the short circuit current strength.

Penalty: Lower gate driving strength.

Question 3: Leakage and Subthreshold-Slope Reasoning [20 points]

Lecture 8 introduces subthreshold conduction and the slope factor. Assume a transistor has subthreshold slope 80 mV/decade. At $V_{GS} = 0V$, its leakage current is $5nA$.

(a) **Leakage increase with gate bias.** Estimate the leakage current if V_{GS} increases from $0V$ to 80 mV, 160 mV, and 240 mV. **[5 points]**

Slope: 80 mV/decade ; at $V_{GS} = 0V \rightarrow I_{leak} = 5nA$

↳ Every increase of 80 mV produces a factor of 10 increase in current.

$0V \rightarrow 80mV$: $I = 5nA(10) = \boxed{50nA}$

$0V \rightarrow 160mV$: $I = 5nA(10)^2 = \boxed{500nA}$

$0V \rightarrow 240mV$: $I = 5nA(10)^3 = \boxed{5000nA}$

(b) **Effect of threshold shift.** A process corner effectively lowers the threshold voltage by 80 mV. Explain why this has approximately the same effect as increasing V_{GS} by 80 mV. Estimate the new leakage current. **[5 points]**

Subthreshold voltage behaviour is: $V_{GS} - V_{th}$

If V_{th} decreases by 80 mV $\therefore V_{GS} - (V_{th} - 80mV) = V_{GS} - V_{th} + 80mV$

which is equivalent to increasing V_{GS} by 80 mV: $V_{GS} + 80mV - V_{th}$ ↑ equiv.

↳ If V_{GS} increases by 80 mV, therefore:

$I_{new\ leak} = 5nA(10)$

$= \boxed{50nA}$

(c) **Sleep-mode leakage.** A block contains 5×10^5 leaking devices. Estimate total leakage power at $V_{DD} = 0.9V$ for the nominal case and for the $-80mV$ threshold-shift case. **[4 points]**

devices : $N = 5 \times 10^5$ devices; $V_{DD} = 0.9V$

- Nominal case : $I_{total\ leak} = 5 \times 10^5 (5nA) = 2.5mA$

  Power : $P_{leak} = VI = 0.9V(2.5mA)$

  $= \boxed{2.25\ mW}$

- -80 mV threshold-shift case : $I_{total\ leak} = 5 \times 10^5 (50 \times 10^{-9}) = 25mA$

  Power : $P_{leak} = VI = 0.9(25mA) = \boxed{22.5\ mW}$

(d) **Performance/leakage tradeoff.** Why might a designer intentionally use lower-threshold devices on a critical path despite the leakage penalty? **[3 points]**

Because lower Vth have greater drive current which reduces gate delay. This is especially beneficial on critical path where delay is the focused not so much the leakage problem.

(e) **Reasoning check.** A student says: "An 80 mV threshold shift is small, so leakage changes only slightly." Use your result to respond. **[3 points]**

Not true. Consider the current question where the given slope is 80 mV/ decade.

This means that every 80 mV reductions produces 10x higher leakage.

We can see this in 3c) where total leakage power increases from 2.25 mW to 22.5 mW with only a 80 mV threshold shift.

Question 4: Interconnect Capacitance, RC Delay, and When Wires Matter [20 points]

A gate drives a wire of length L. The wire has resistance per unit length $0.08\Omega/\mu m$ and capacitance per unit length $0.20fF/\mu m$. The driver has intrinsic gate delay $35ps$.

(a) **Wire delay for three lengths.** Compute the wire delay for:

$$L = 500\mu m, \qquad 2mm, \qquad 5mm.$$

Use consistent units and clearly show your reasoning. **[6 points]**

Uniform distributed RC wire: $t_{wire} = \frac{1}{2} R_{wire} C_{wire} = \frac{1}{2} rL\,cL$

- $L = 500\mu m$: $R = rL = 0.08(500) = 40\Omega$, $C = cL = 0.2(500) = 100fF$ $\rightarrow$ $t = \frac{1}{2}(40)(100\times10^{-15}) = \boxed{2ps}$
- $L = 2mm$: $R = rL = 0.08(2000) = 160\Omega$, $C = cL = 0.2(2000) = 400fF$ $\rightarrow$ $t = \frac{1}{2}(160)(400\times10^{-15}) = \boxed{32ps}$
- $L = 5mm$: $R = rL = 0.08(5000) = 400\Omega$, $C = cL = 0.2(5000) = 1000fF$ $\rightarrow$ $t = \frac{1}{2}(400)(1000\times10^{-15}) = \boxed{200ps}$

(b) **Critical length.** Find the wire length at which the wire delay becomes equal to the intrinsic gate delay. **[4 points]**

$t_{wire} = t_{gate} = t_{intrinsic} = 35ps$

$t_{wire} = \frac{1}{2} RC = \frac{1}{2}(rL)(cL) = \frac{1}{2} rcL^2$

$\therefore 35 = \frac{1}{2} rcL^2$

$35 = \frac{1}{2}(0.08)(0.2\times10^{-15})L^2$

$\Rightarrow L \approx \boxed{2.09\ mm}$

(c) **Capacitance and dynamic power.** For each of the three lengths in part (a), compute the wire capacitance. Assuming $V_{DD} = 0.9V$, activity factor 0.1, and frequency 1 GHz, compute dynamic power due to the wire capacitance. **[4 points]**

$P_{dyn} = \alpha C V_{DD}^2 f$; $VDD = 0.9V$; $\alpha = 0.1$; $f = 1GHz$

C from a)

- 500 µm : $P = 0.1(100 \times 10^{-15})(0.9)^2(1 \times 10^9) =$ **8.1 µW**
- 2mm : $P = 0.1(400 \times 10^{-15})(0.9)^2(1 \times 10^9) =$ **32.4 µW**
- 5mm : $P = 0.1(1 \times 10^{-12})(0.9)^2(1 \times 10^9) =$ **81 µW**

(d) **Local versus global interconnect.** Explain why local interconnect and global interconnect scale differently as technologies scale. Your answer should refer to the conceptual distinction between local interconnect scaling with technology and global interconnect scaling with die size. **[3 points]**

- Local interconnect : Connect nearby transistors so its physical length decreases as technology dimensions get smaller.
- Global interconnect : Takes up a larger portion of the chip so its length depends on overall die size & not on transistor technologies scaling.

∴ Global interconnect does not get proportionally smaller as transistor dimensions shrink.

(e) **Design action.** For the 5 mm wire, propose two fixes to reduce delay. Examples include repeater insertion, wider wire, using a higher metal layer, or architectural floorplanning. For each fix, state one tradeoff. **[3 points]**

1) Repeater insertion : Break a long wire into several shorter wire segments with buffers.
Bc $t_{wire} \propto L^2$ ∴ shorter wire → lower delays.
Tradeoff : Increases in area, capacitance & power.

2) Wider wire : Wire resistance is $R = \rho \frac{L}{A}$ so increasing width (cross-section area) reduces Resistance.
↓ bc $t \propto RC$ ∴ lower delay
Tradeoff : More materials & costs, can also increase capacitance.

Question 5: Static CMOS, Ratioed Logic, and Pseudo-NMOS Design Tradeoffs [20 points]

Consider implementing the logic function:

$$Y = \overline{A + B + C + D}$$

with two alternative styles:

Option 1: Static complementary CMOS 4-input NOR.

Option 2: Pseudo-NMOS NOR with one always-on PMOS pull-up load and four NMOS pull-down devices in parallel.

Assume $V_{DD} = 1.0V$. For the pseudo-NMOS gate, the effective pull-up resistance is $30k\Omega$. Each ON NMOS pull-down has resistance $10k\Omega$. The output load capacitance is $40fF$.

For the static CMOS NOR, assume the worst-case pull-up resistance is $40k\Omega$, and the worst-case pull-down resistance is $10k\Omega$.

(a) **Logic levels for pseudo-NMOS.** Compute V_{OL} for the pseudo-NMOS gate when exactly one NMOS pull-down is ON. Then compute V_{OL} when all four NMOS pull-down devices are ON. Explain why V_{OL} is input-pattern dependent. **[5 points]**

$$V_{OL} = V_{DD}\frac{R_N}{R_p + R_N}$$

Legend: R_N : nMOS resistance
R_p : pMOS resistance

- 1 nMOS is ON : $R_N = 10k\Omega$

$$\therefore V_{OL} = 1\left(\frac{10}{30+10}\right) = \frac{10}{40} = \boxed{0.25\,V}$$

- All 4 nMOS are ON : $R_{N4} = \frac{R_N}{4}$ (parallel)

$$= \frac{10}{4} = 2.5\,k\Omega$$

$$\therefore V_{OL} = 1\left(\frac{2.5}{30+2.5}\right) = \boxed{0.0769\,V}$$

V_{OL} is input pattern dependent because nMOS are in parallel meaning different input → diff # of nMOS turning ON

(b) **Static power in pseudo-NMOS.** Compute static power in the pseudo-NMOS gate for the one-pull-down-ON case and the four-pull-down-ON case. **[4 points]**

$I_{DC} = \frac{VDD}{R_p + R_N}$; $P_{static} = V_{DD} I_{DC}$

- 1 nMOS ON : $R_{total} = 30 + 10 = 40\,k\Omega$; $I_{DC} = \frac{1\,V}{40\,k\Omega} = 25\,\mu A$

$\therefore P_{static} = (1\,V)(25\,\mu A) = \boxed{25\,\mu W}$

- 4 nMOS ON : $R_{total} = 30 + 2.5 = 32.5\,k\Omega$; $I_{DC} = \frac{1V}{32.5k\Omega} = 30.77\,\mu A$

$\therefore P_{static} = (1\,V)(30.77\,\mu A) = \boxed{30.8\,\mu W}$

maybe clearer to be more explicit.

(c) **Delay comparison.** Estimate the falling-output delay for both designs when one NMOS pull-down is ON. Estimate the rising-output delay for both designs. Clearly state what resistance you use for each case. **[5 points]**

$t_p = \ln(2) R_{eq} C_L$; $C_L = 40fF$

- static : Falling (1→0)

1 nMOS : $R = 10k\Omega \rightarrow t_{p_{HL}} = \ln(2)(10k\Omega)(40fF)$

$\approx \boxed{277\,ps}$

Rising (0→1) : $R = 40k\Omega$ (worst-case) $\rightarrow t_{p_{LH}} = \ln 2 (40k\Omega)(40fF)$

$\approx \boxed{1.11\,ns}$

- pseudo-nMOS Rising (0→1) so nMOS is OFF : $R = 30k\Omega$ ↳ always ON pMOS

$t_{p_{LH}} = \ln(2)(30k\Omega)(40fF) \approx \boxed{832\,ps}$

Falling (1→0) so 1 nMOS is ON : $R = 10k\Omega \rightarrow t_{p_{HL}} = \ln(2)(10k\Omega)(40fF) = \boxed{277\,ps}$

(d) **Design tradeoff.** Which implementation would you choose for a low-power design? Which might be attractive for reduced transistor count or input capacitance? Explain. **[4 points]**

- Low power:
  - Static CMOS
    - Bc no direct path from VDD → GND so very small power.

  For pseudo-nMOS, it does have static current if output is low (0).

- Reduced transistor count / Input Cap
  - Pseudo nMOS

: static → 4pMOS + 4nMOS = 8 transistors

pseudo → 1pMOS + 4nMOS = 5 transistors

(e) **Beyond fan-in four.** Suppose the function becomes:

$$Y = \overline{A + B + C + D + E + F}.$$

Explain why directly implementing this as a single high-fan-in gate is unattractive, and propose a better decomposition. **[2 points]**

Bc it requires 6 pMOS in series for the pull up network

∴ $R_{eq} = \sum^{6} R_p$; bc $t \propto RC$ so delay also becomes large.

Better: Idea → Break into smaller fan-in stages.

$$X_1 = \overline{A + B + C}$$
$$X_2 = \overline{D + E + F}$$
$$\Rightarrow Y = X_1 X_2$$
$$= \left[\overline{A + B + C}\right]\left[\overline{D + E + F}\right]$$
$$= \overline{A + B + C + D + E + F}$$

other equivalent implementation also works

Question 6: Complete the Cadence Virtuoso Tutorial 1 posted on Blackboard.